A Seminar Presentation
On

RYDBERG CONSTANT

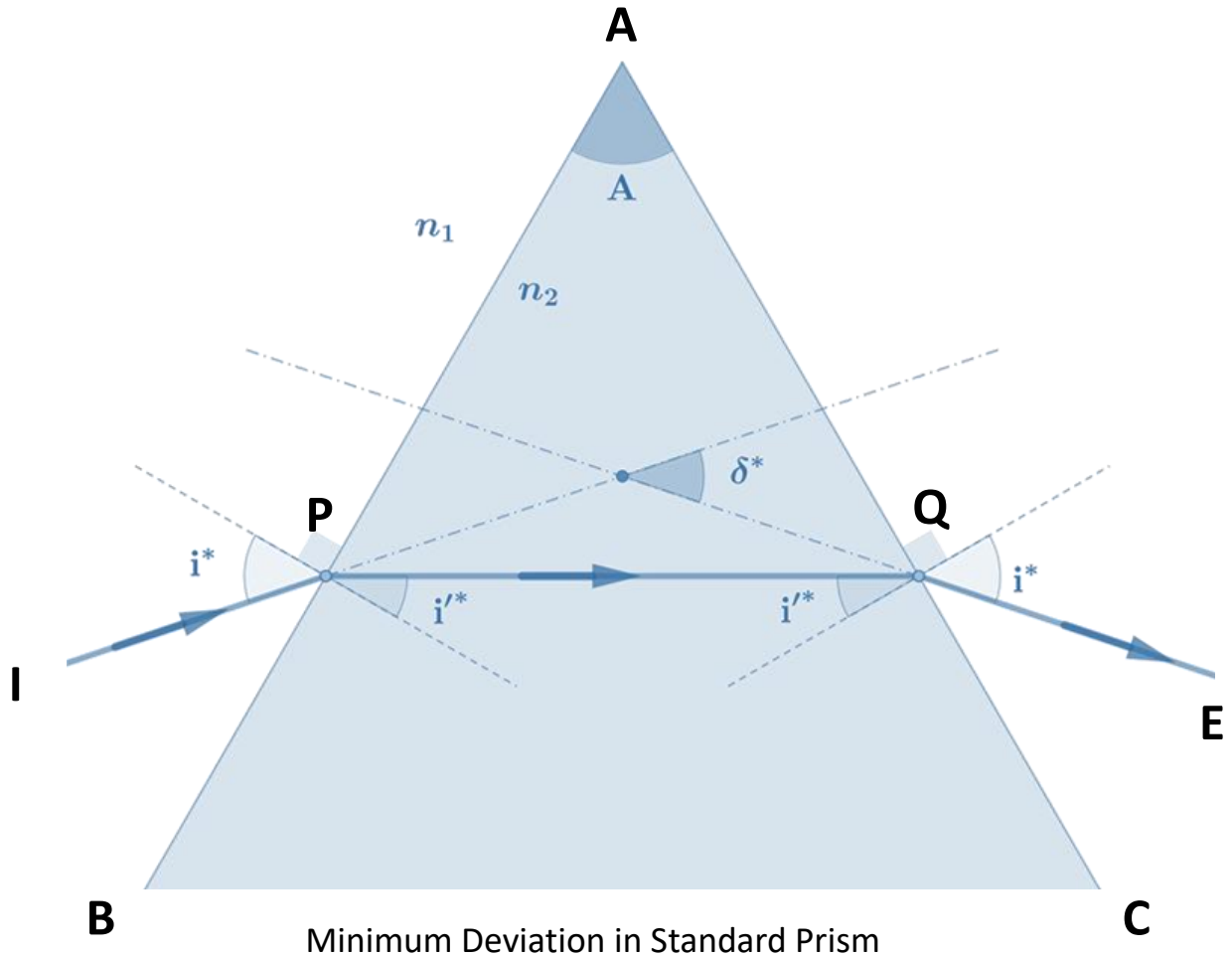
By Kumbhar Kaustubh Rajshekhar

Aim:

To measure the wavelength of the Balmer lines of Hydrogen and hence to determine the Rydberg constant.

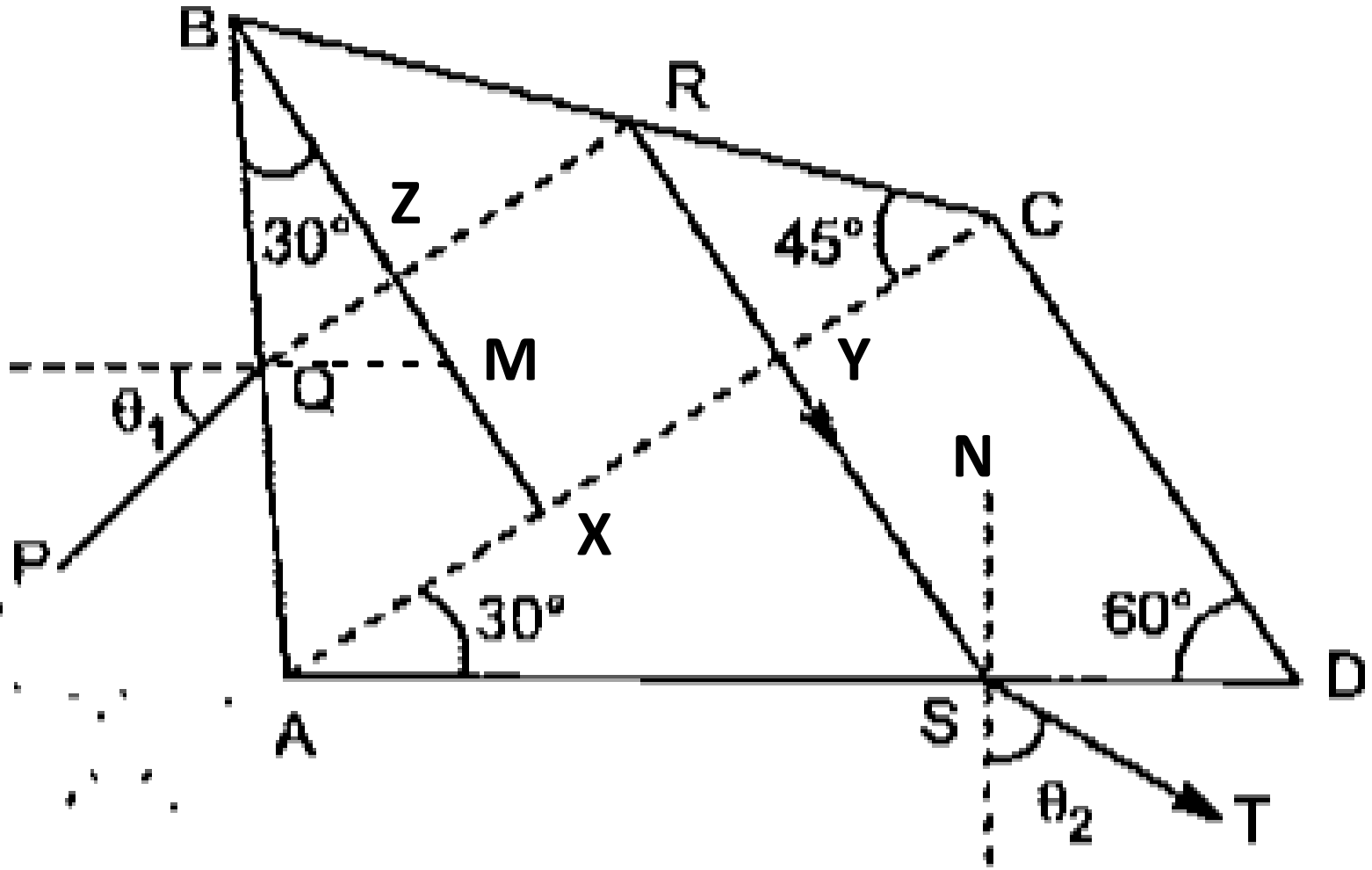
Apparatus:

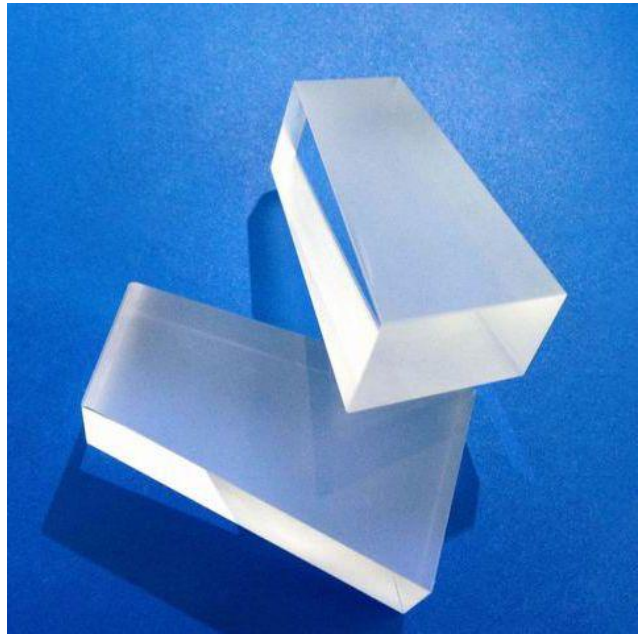
Constant deviation prism, constant deviation spectrometer, Mercury source, Hydrogen source, etc.



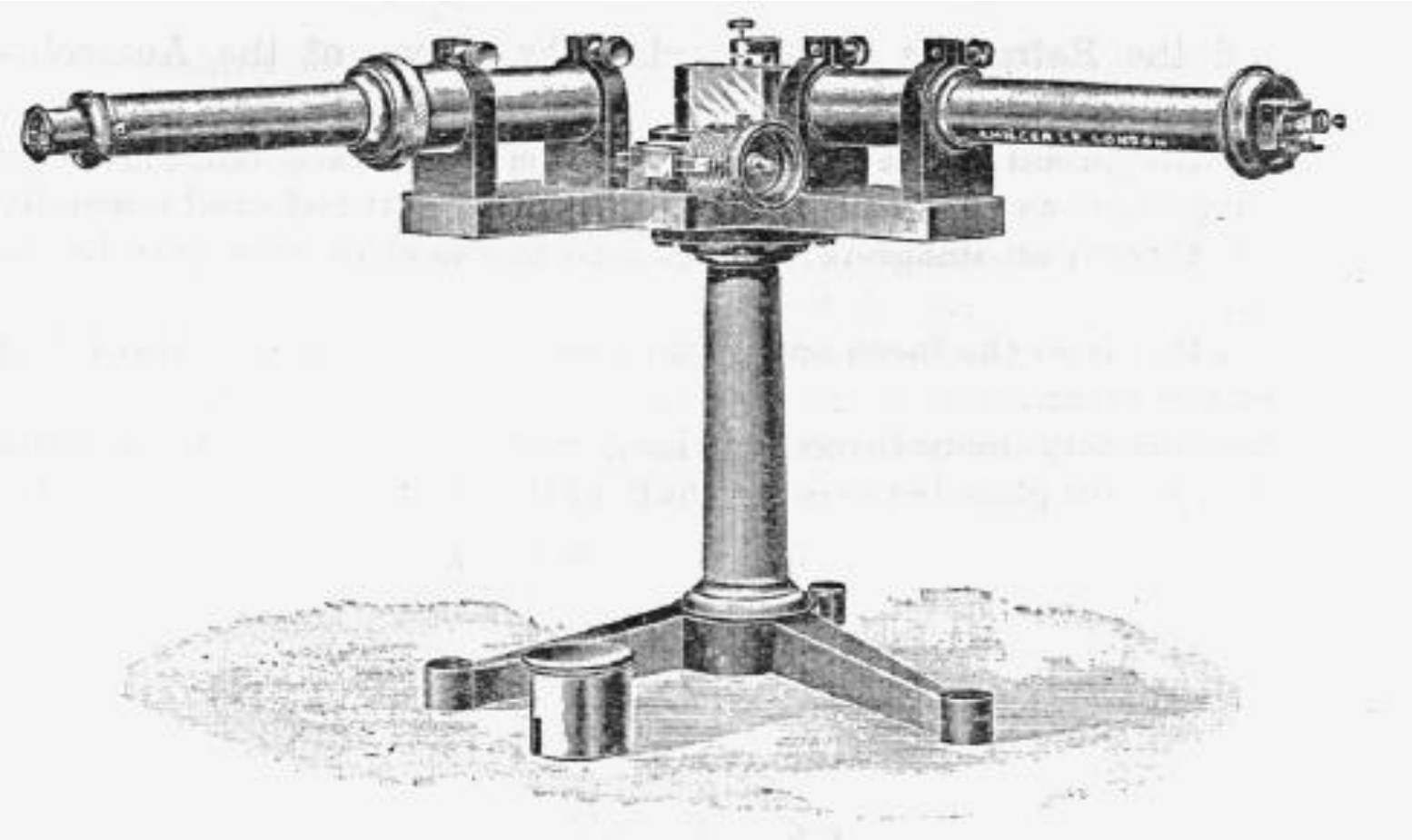
Minimum Deviation in Standard Prism

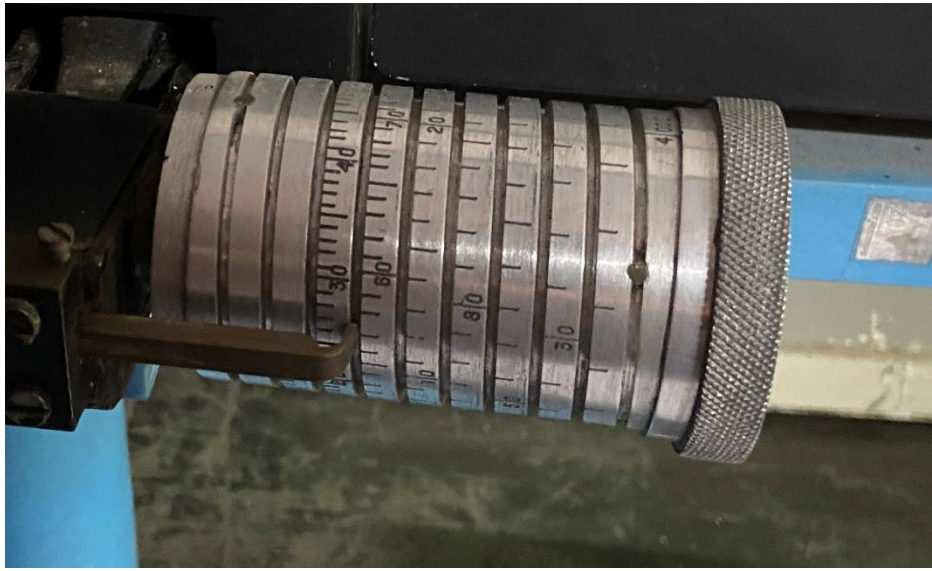
CD Prism



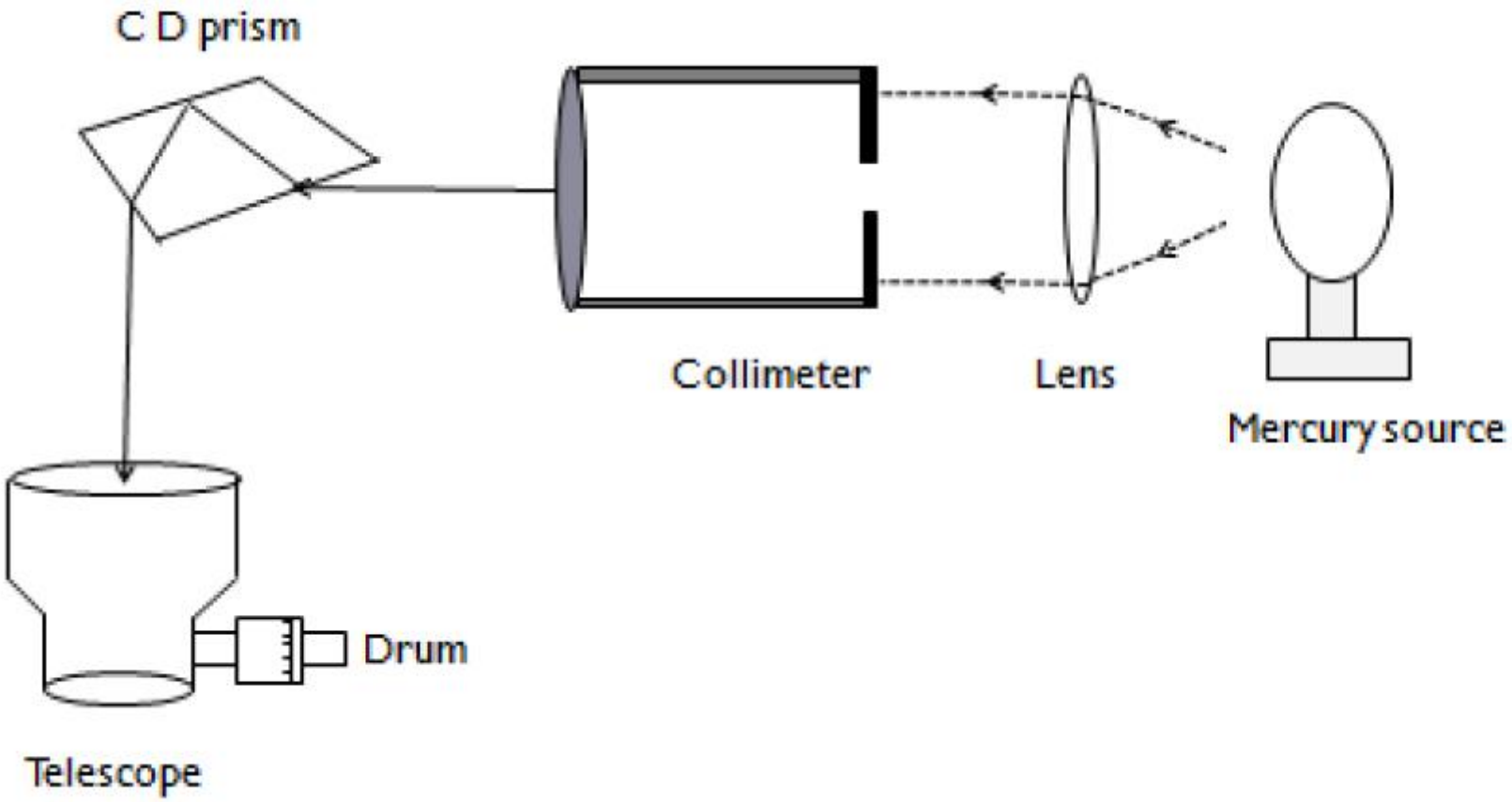


CD Spectrometer & Drum





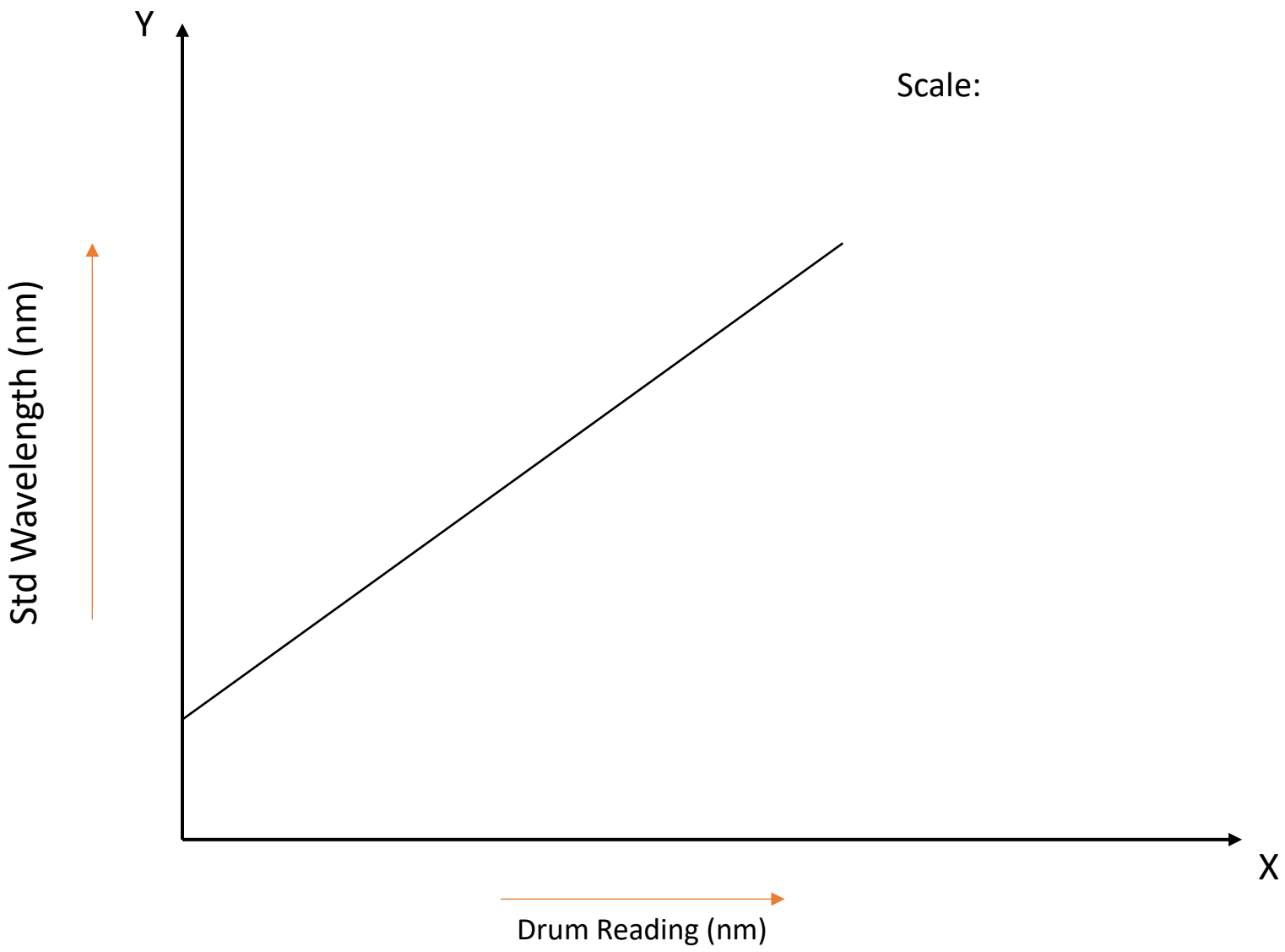
Calibration of CD Spectrometer



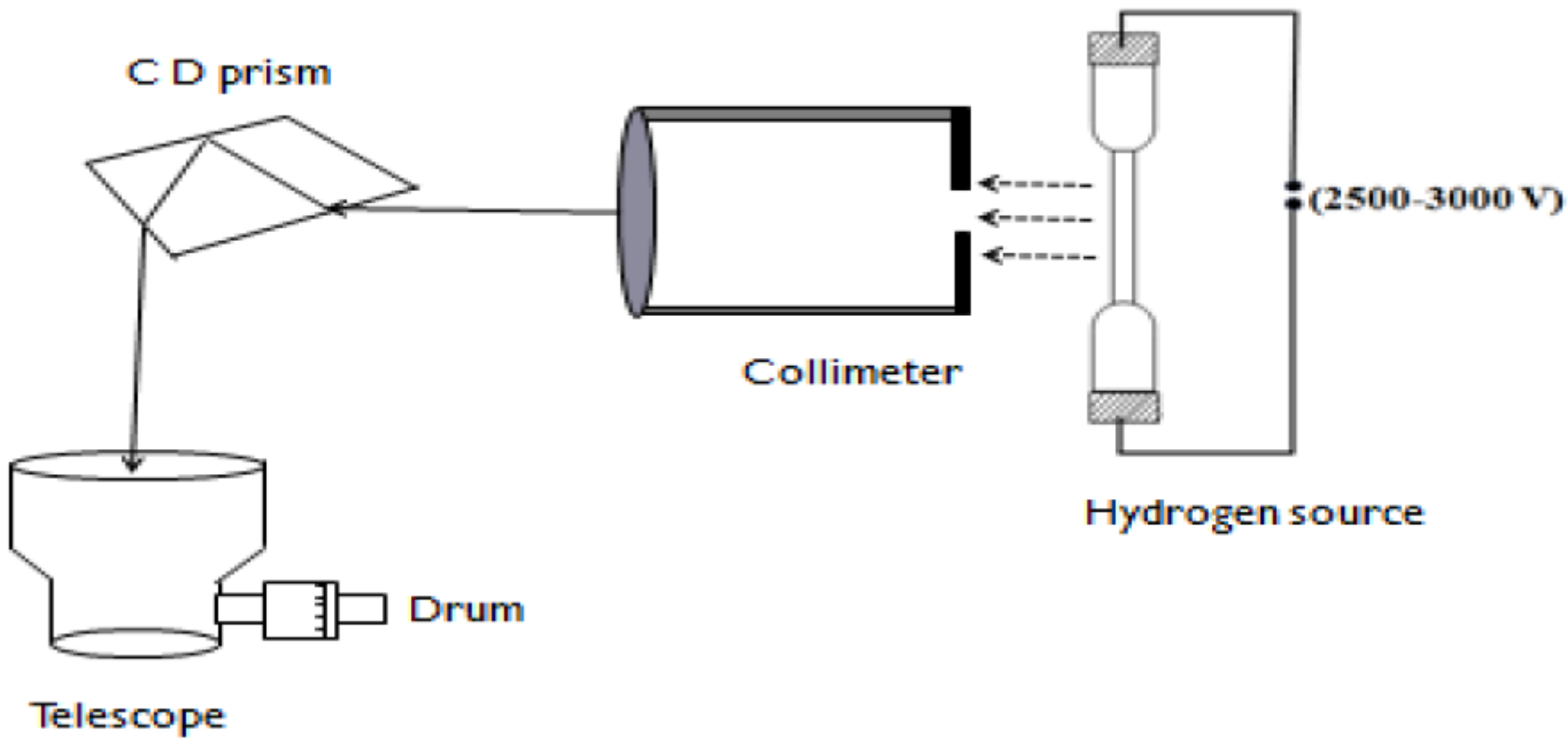
Part 1- Calibration of Spectrometer

Obs No	**Spectral Lines**	**Drum Readings (nm)**	**Standard Wavelengths (nm)**
1	Red I		643
2	Red II		636.5
3	Yellow I		597.8
4	Yellow II		593.2
5	Green		559.6
6	Violet		410.5

Drum Reading vs Standard Wavelength



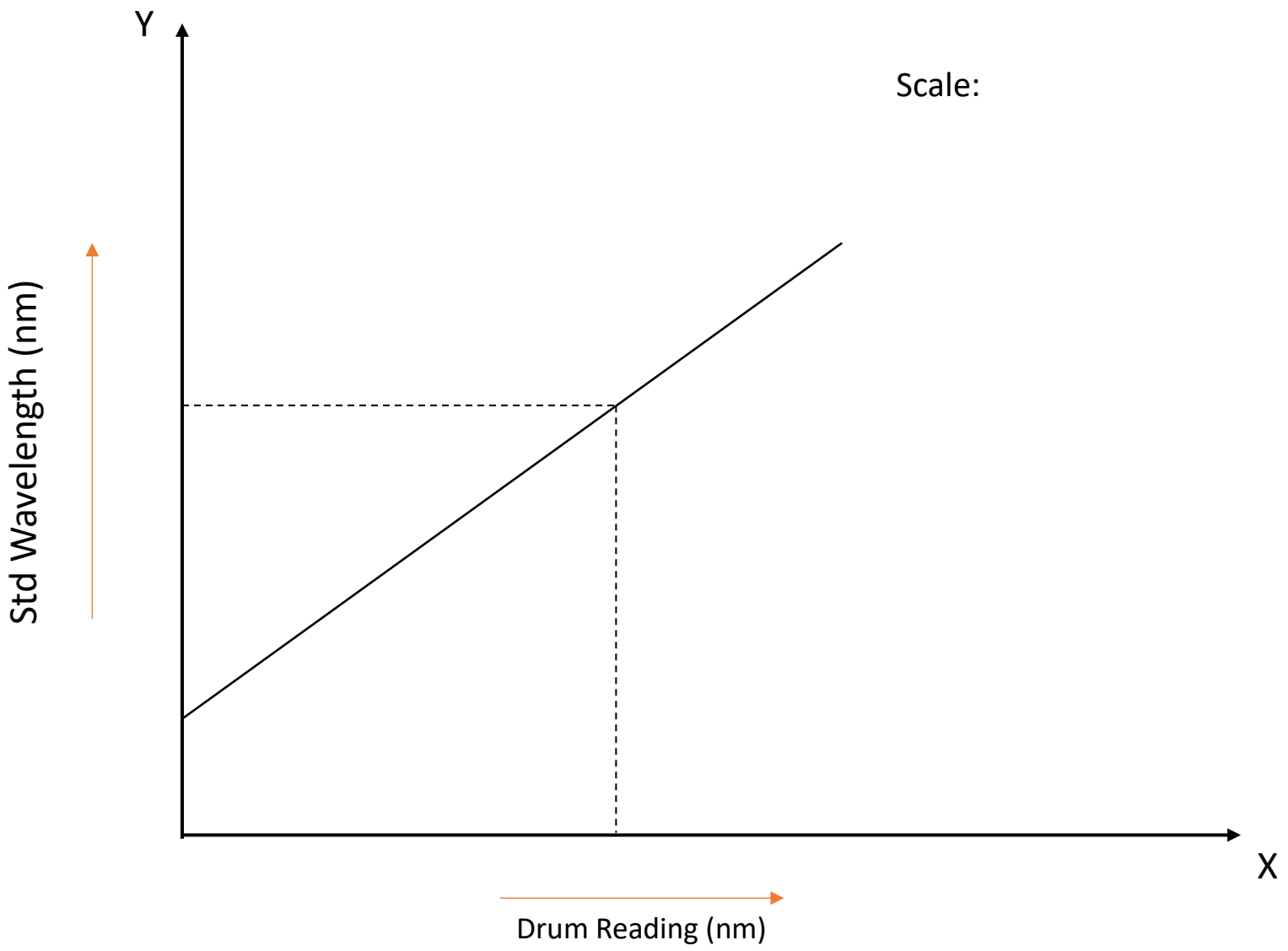
Part 2 - Measurement of Balmer Series

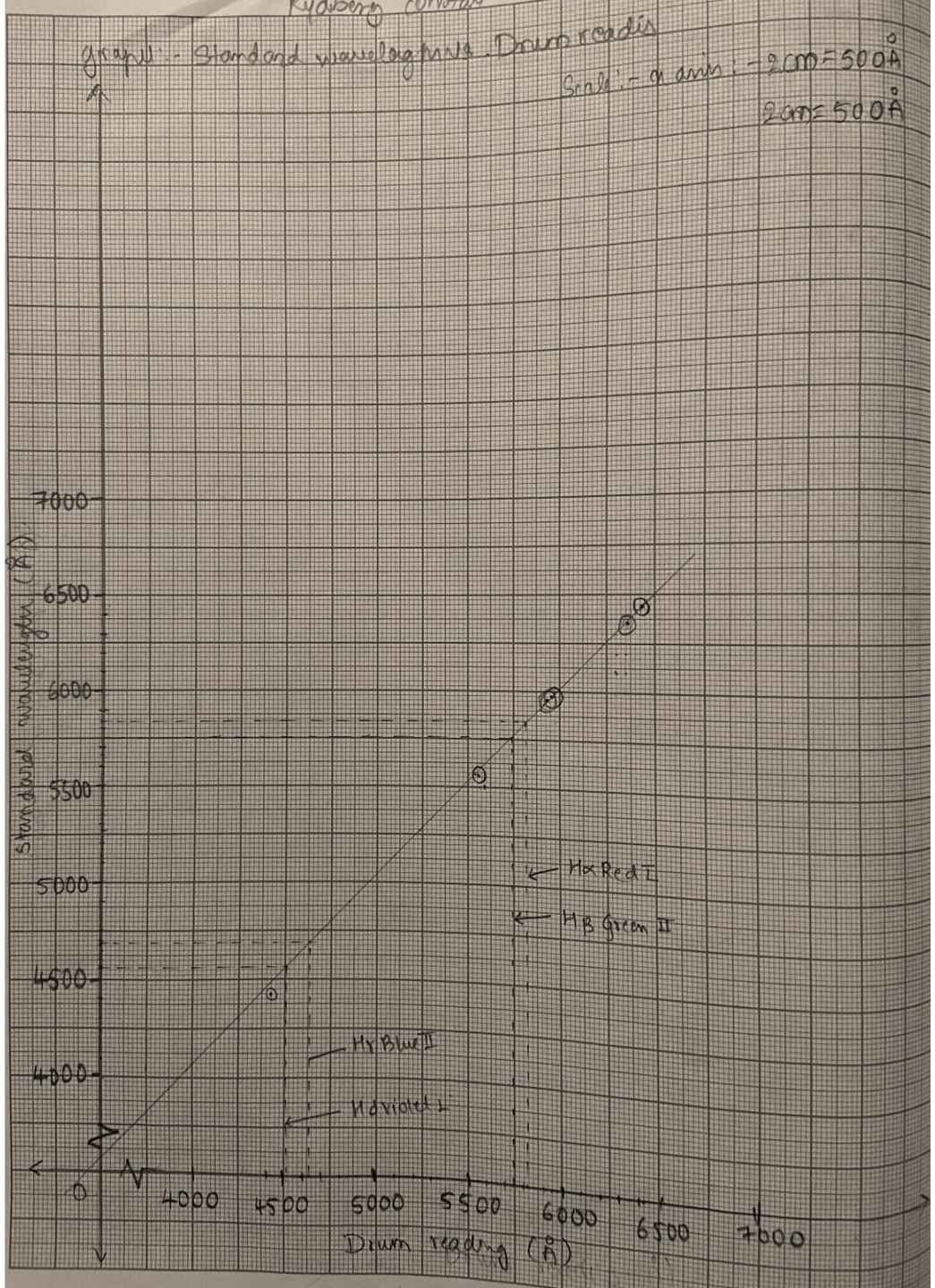


Part 2 - Measurement of Balmer Series

Obs No	Spectral Lines	Drum Readings (nm)	Corrected Drum Readings (nm)	Rydberg Constant (m^{-1})
1	H_{α} (Red I)			
2	H_{β} (Green II)			
3	H_{γ} (Blue II)			
4	H_{δ} (Violet)			

Drum Reading vs Standard Wavelength




graph :- Standard wavelength vs Drum readis
Scale :- x axis :- 2cm = 500Å
2cm = 500Å
Standard wavelength (Å)
7000
6500
6000
5500
5000
4500
4000
Hα Red I
Hβ Green II
Hγ Blue II
Hδ violet I
0
4000
4500
5000
5500
6000
6500
7000
Drum reading (Å)

Formula:

$$\frac{1}{\lambda} = R\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$$

Standard wavelengths of Balmer lines :

H_α = 6562.10 Å

H_β = 4860.74 Å

H_γ = 4340.10 Å

H_δ= 4101.20 Å

Results:

1. R_{α} =

2. R_{β} =

3. R_{γ}=

4. R_{δ} =

5.R_{mean} =

Conclusion: The standard wavelength of Balmer series and observed wavelength Balmer series are nearly equal.

Theory

Rydberg Constant is one of the most accurate and well determined constants. Each element (of periodic table) has its unique Rydberg constant. It can be determined using

$$R_{\infty} = \frac{m_e e^4}{8 \varepsilon_0^2 h^3 c}$$

$$= \frac{(9.11*10^{-31})(1.6*10^{-19})^4}{8\,(8.85*10^{-12})^2\,(6.626*10^{-34})^3\,(3*10^8)}$$

$$= 1.097*10^7 \ m^{-1}$$

Postulates of Bohr Atomic Model

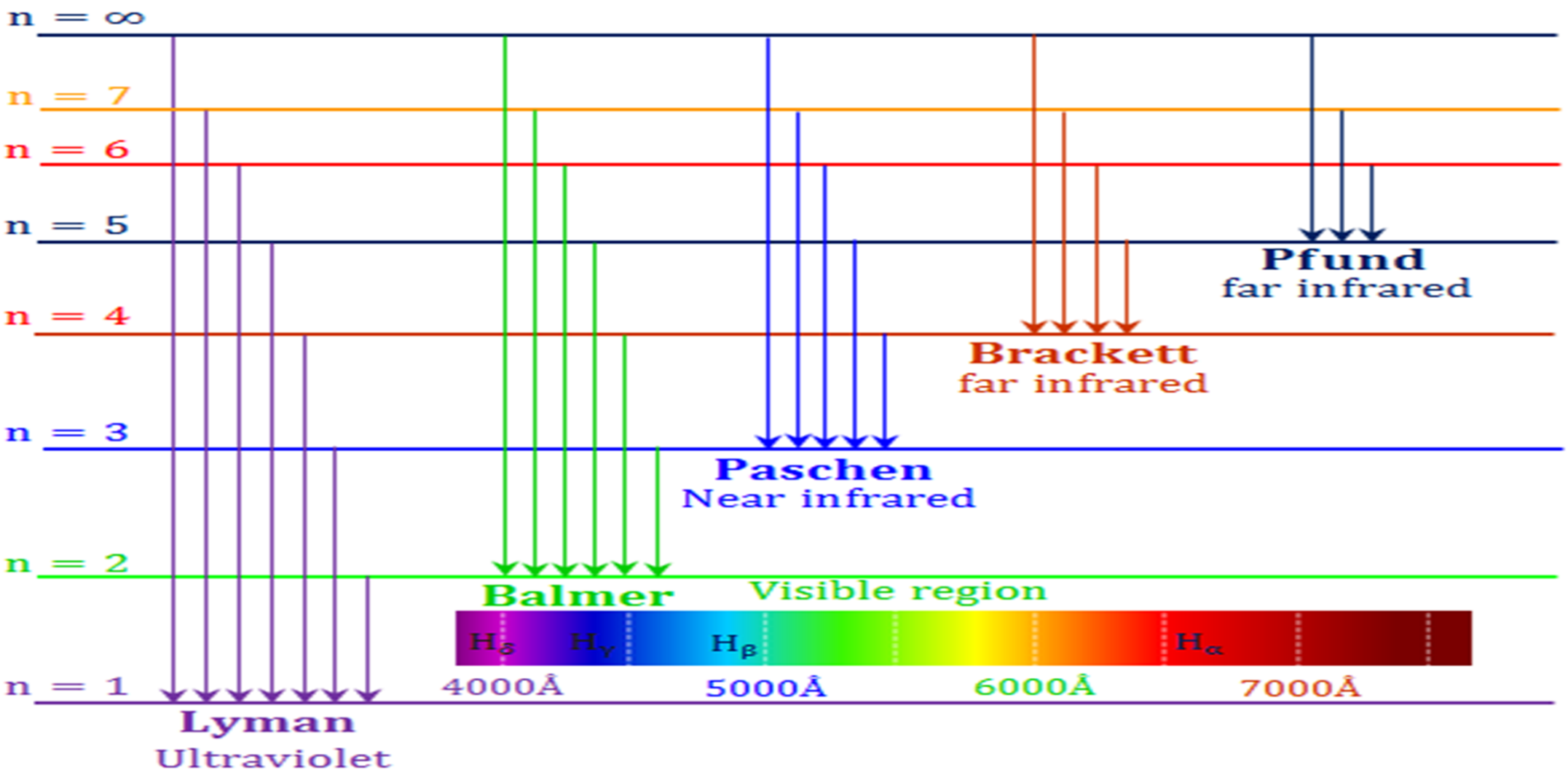
1) Electrons revolve around the nucleus in a fixed circular path termed “orbits” or “shells” or “energy level.”

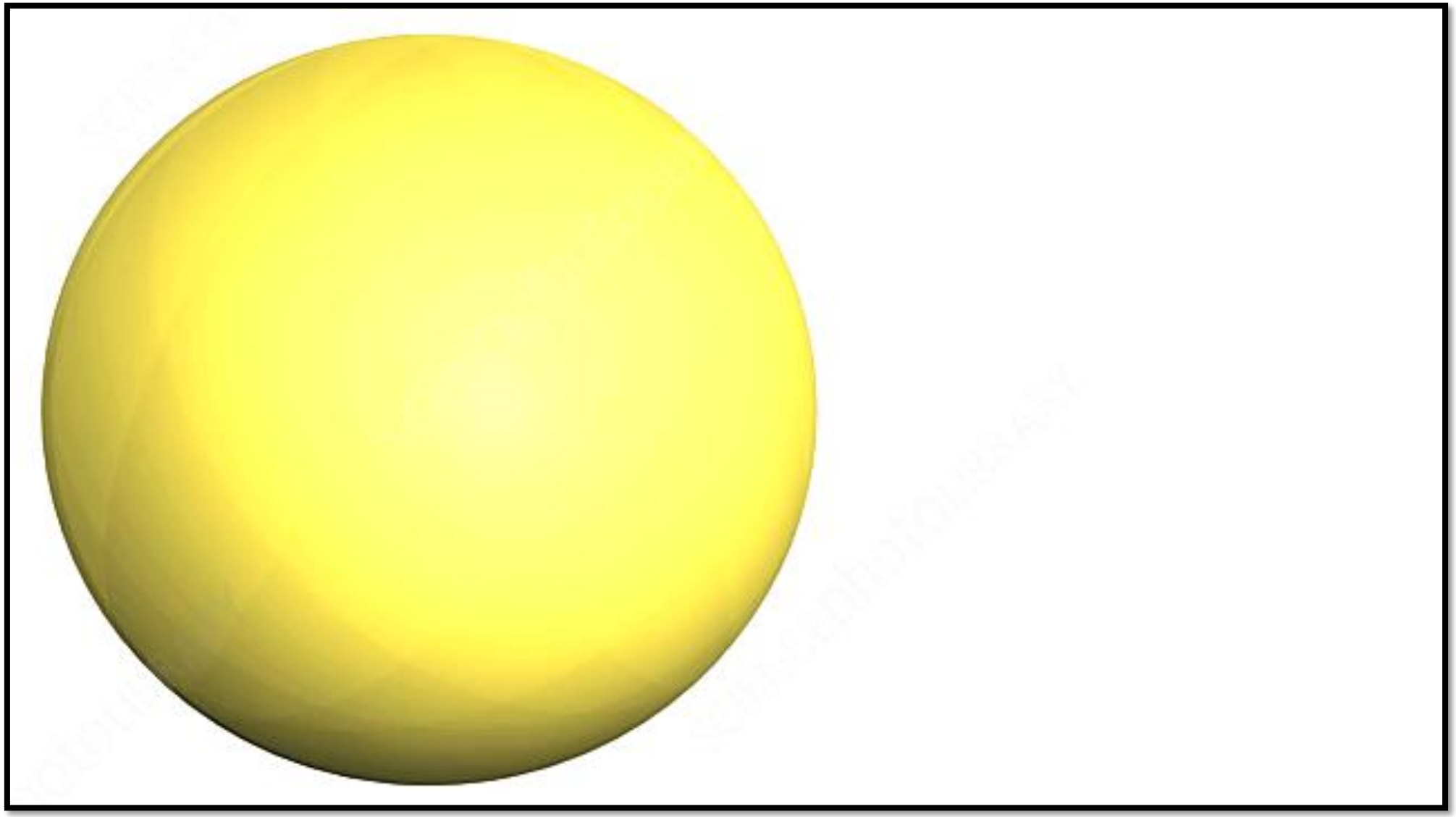
2) The radius of the orbit of an electron can only take certain fixed values such that the angular momentum of the electron in these orbits is an integral multiple of h/2Π.

3) The electrons in an atom move from a lower energy level to a higher energy level by gaining the required energy and an electron moves from a higher energy level to lower energy level by losing energy. It loses its energy by emitting a photon.

$\Delta E = h\nu$

All the spectral lines observed in the **Hydrogen spectrum** can be classified into different series: **Lyman**, ***Balmer***, **Paschen**, **Brackett** and **Pfund series**.





Applications :

1) To calculate wavelength and wavenumber
2) To calculate the energy of shells
3) To study the atomic spectra

Thank You